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CONTEXTUAL DESIGN

Karen Holtzblatt
Incontext Enterprises

INTRODUCTION

A Brief History of Contextual Design

In the mid-1980s, the industry was looking for ways to make products more usable. Usability as a field was getting underway, testing products in labs and providing a feedback loop at the end of the design process. However, as John Whiteside at Digital Equipment Corporation had predicted, usability testing was only bringing a 15–20% improvement to user experience. It could not address the definition of product function or significantly change the structure of the product because it occurred at the end of the design process. The challenge was how to redress this and really impact the way products fit into people's lives.

Contextual Inquiry, and subsequently Contextual Design, was developed and shaped in response to that challenge. This was an exciting but demanding time, when many in the industry were trying to move engineering-driven organizations to use customer data at the beginning of the product development cycle. Central to this discussion was the question of what kind of data would best drive product concept design and structure.

Engineering groups recognized that the typical marketing data gathered in surveys, focus groups, and user group conferences could target problem areas for customers, but could not give them the detailed data that they needed about how people worked. They implicitly knew that to really define products—to understand what functionality is needed by people and how to structure that functionality within the product to best support work—required a deep understanding of the blow-by-blow details of what people did to get their jobs done. But this information could not be collected with surveys or typical marketing interviews. Marketing techniques provided neither the detail about how the product fit into an overall work practice that involved other tools and manual processes, nor a coherent picture of the user's intent and tasks to guide the structuring of function within the product. Organizationally, this resulted in conflict between marketing, engineering, and usability groups over what to make and how to structure it.

Contextual Inquiry, along with other techniques for going into the field to see how work practice unfolds, was a response to this need. The problem was not that groups could not get along. The problem was that the data needed for design was simply not being collected. To make products more usable, to make products that people really wanted and could use meant understanding what people were really trying to do and designing new technology to support, extend, and transform that practice. But without knowing the practice, missing the target or breaking the practice was probable.

Although initially resisted, by 1990 the idea of user-centered design had spread to the point that people wanted to design with an understanding of work practice. Within companies, evangelists were pushing for site visits, field data collection, and so forth. By the mid-1990s, people were collecting volumes of work practice data and looking for ways to handle it, and the practices of iterative design, rapid prototyping, participatory design with the customer, work modeling, and field research were evolving in parallel. The practices of Contextual Design, the full customer-centered front-end design process incorporating Contextual Inquiry, were designed in support of these changes over the last 13 years by working with teams in companies throughout the industry.

Today, the ideas of user-centered design, user experience, and designing from customer field data have become the norm. Not all companies use all steps of Contextual Design or have good front-end, user-centered design processes. However, the industry has turned the corner from resistance to awareness. Customer-centered design is being pushed in organizations by management and designers alike; people are seeking skills, universities are teaching customer-centered processes, and companies are creating user-centered design methodology groups.

The idea of user-centered design has become central to the consciousness of people who produce products or systems, marking a successful industry change. But, at the level of the individual organization, change is often just beginning. This chapter provides an overview of the steps of Contextual Design (Beyer & Holtzblatt, 1998) and also highlights lessons learned over 13 years spent helping organizations adopt and tailor Contextual Design for organizational use.

Contextual Design Overview

Contextual Design is a full front-end design process that takes a cross-functional team from collecting data about users in the field, through interpretation and consolidation of that data, to the design of product concepts and a tested product structure (see Table 49.1 for the steps). The strength of Contextual Design is that it tells people what to do at each point so that they can move smoothly through the design process from customer data to specific interaction design and code. Contextual Design is often referred to as good scaffolding because it works like a backbone to which other tools and techniques can be easily added. For example, to allow more creative thinking, add additional brainstorming processes before visioning; to incorporate classic market studies, use the vision to structure questionnaires; or to do use cases, add them after the user environment design has stabilized. Also, Contextual Design can be viewed as series of techniques that can be incorporated into a company's standard methodology—it produces artifacts and data that can feed existing requirements specification formats.

Contextual Design therefore can be easily tailored to the needs of companies and adopted one technique at a time. Organizational adoption, which is discussed more at the end of this chapter, depends both on the readiness of people to try something new and the skill with which people alter and incorporate Contextual Design techniques. Piecemeal adoption and alteration of the method to fit the company is a normal part of organizational change. All introduction of user-centered design techniques is about organizational change.

When companies introduce customer-centered design into their organizations, the adoption will change the process itself. Changing the process requires making trade-offs and decisions about what to keep, what to change, and how. However, because each technique serves a specific intent, Contextual Design practitioners need to understand the intention(s) of each step,

TABLE 49.1. The Steps of Contextual Design

Step	Description	Intent
Contextual Inquiry	Collect data using ethnographic techniques by observing and questioning customers while they work.	Get reliable knowledge about what people do, what they care about, and the structure of the work practice.
Interpretation Session and Work Modeling	Capture the key issues of one individual's work practice and model their work using the five work models, each capturing a different dimension of work practice.	Create a multidimensional understanding of the customer shared among cross-functional groups who have to cooperate in the design. Provide a way of capturing complex qualitative data.
Affinity and Work Model Consolidation	Consolidate the individual work models and issues to reveal the structure of the work across a population without losing individual variation.	Create coherent reusable data about a customer population to guide conversations between designers identifying key requirements and stimulating design discussions.
Visioning and Storyboarding	Invent how new technology will address the user work practice by creating a high-level story (the vision) of how work will be changed. Work out the details of this story at the level of the task (storyboard).	Ensure that the technology will fit into the overall work of the user. Guide invention with data at all levels to avoid breaking the work and ensure supporting the work is supported with technology.
User Environment Design	Design a floor plan that shows how the parts of the new system will interrelate. Represent the structure and function clustering of the system independently of consideration of User Interface and implementation.	Ensure that the flow of work within the system the supports the work. Help designers focus on work flow and function within the system instead of considerations of the User Interface and implementation. Create a system representation to support planning.
Paper Prototyping	Test and modify the new system design in partnership with customers using paper mock-ups of the User Interface. Have people do real work tasks in the paper system, not just review it and provide opinions.	Provide a reliable way of talking with users about the proposed system. Verify the design before committing ideas to code. Deepen the requirements of the system. Start testing User Interface ideas and product concepts before shipping.

why the step is included, and what is gained or lost through trade-offs or changes.

For example, central to any customer-centered design process is the collection of rich, reliable data about how people work. People often ask why they cannot work from the "collective" experience of marketers, sales people, business analysts, and others who have years of experience, and then apply the data organization and visioning techniques to that data. But this violates the most basic value of Contextual Design—design from reliable data collected in an agreed-upon way that represents the actual things that people do. As long as the data is coming from someone's opinion—however much experience behind it—its validity and quality are arguable. To create a data-driven organization, the quality and reliability of the customer data must be unquestioned. Challenging this requirement drives teams back into the age-old argument about whose experience is dominant when conflict arises. The Contextual Design methods of field data collection and consolidation ensure that the data is good and the rules of interpreting it clear. Real customer data ensures that team members can create a shared understanding about the data and requirements across business functions. So Contextual Design teams never trade off getting real data from the field to guide the design. As a result, organizations learn to make decisions based on data and to seek data to make decisions. This alone has enormous value within the product development process.

Similarly, each step in Contextual Design was created for a purpose and is multidetermined; there are a number of reasons why and how the exact steps have been assembled. When adopting or redesigning Contextual Design, first understand the fundamental intents of the steps and the parts. When truncating a step, leaving out a part, or making a change, make sure that the fundamental intent of that step is not violated without purpose. The next section provides an overview of each step (see Table 49.1) identifying its fundamental intent and discussing reasonable trade-offs.

Setting Project Scope

A successful project and successful organizational adoption is as much about defining the project scope as it is about organizational readiness. Organizations and teams are always in the middle of delivering something. Asking people to slow down to do something new or different will always be difficult. Even for people with the best intentions, figuring out how to insert customer-centered processes into an existing organization is daunting. Resistance shows up as questions about the length of time this new process will take, a concern that old knowledge will be ignored and a fear that schedules will slip.

One way to address these concerns is to define the project scope in a way that works for the organization. Because Contextual Design is a set of techniques, it can be tailored in response to issues of time, resources, and place in the life cycle.

The first step of any project is to set the project scope. This involves deciding how wide to open the inquiry and how much change the organization and its people are ready to make. A wide scope requires the team to collect more data, make more models, and create more designs than a small scope. For example, a project to understand how a Web site is performing as a sales site could involve a process to watch eight people interact with that site, identify the top key issues, and generate some design ideas and quick fixes to improve the site. But a project to define how buying and selling could be reinvented by the web will involve interviewing many people in different contexts: Looking at the sites of multiple companies, observing how people buy in stores and from catalogs, understanding how families make decisions collaboratively, and determining what influences buying decisions.

The wider the scope, the longer the process and the more resources required. The narrower the scope, the shorter the process and the fewer the resources required. A wide scope will produce more change to an existing design. Change to an existing plan is always threatening because it means changing direction and writing more code. The scope of the project will determine the process, but it also must fit corporate goals and peoples' willingness to make change.

Defining the Right Process for the Project

The question for process designers is not whether teams should use Contextual Design, but what teams should do given their goals. A number of factors influence the length of the project: How easy or difficult it is to set up the customer visits, the number of users to be interviewed, the number of work models to be consolidated, the complexity of the design, and the number of people available to do the work. The time to communicate to others and to obtain organizational buy-in will also impact time. Planning the project process means planning all these things, but project scope will affect them all. Consider some of the following differences in process influenced by changing the project scope:

- **What are the top 10 issues I could fix in my existing project?** Talk to 6–8 customers about the current product and its competition, and watch customers using the product. In the interpretation session, capture only affinity notes, not the models. Create the affinity and identify the top issues. This can be done in two weeks. Once the team identifies the top issues, they can use their normal processes to generate new ideas. Or, they can generate new ideas and then mock them up and iterate them in paper prototypes interviews.
- **How is my design, which is already planned?** If the requirements, the vision, the structure, and user interface drawings are already done, no one wants to start from the beginning. Do a Reverse User Environment Design from the user interface drawings and examine the proposed structure. Fix the structure based on User Environment Design principles, make new paper prototypes, and iterate them with 8–10 customers. This gets customer data into the process without asking anyone to start over.
- **What can I do for a next release or stepwise change?** If the company is not open to significant change for the next release, look at the product, its competition, and manual processes. Manual processes may give ideas on what can be improved. Interview three people for each of three significant job titles using the product. Capture affinity notes, sequences, and artifacts. Consolidate only the affinity and sequences. Then vision new function and storyboard using the sequences to drive the storyboard. Build a Reverse User Environment Design of the product and integrate new function into the existing User Environment Design. Paper prototype and test it with the new system.
- **When do I need to use the full process?** It is only when defining a new product, a major change to a product, opening a new market, or understanding how to use new technology that the full Contextual Design process and all the models are necessary. The full process will push design thinking out much more broadly and bring new significant functionality into the design.

Contextual Design can handle very large scoped projects—the models naturally extend the breadth of ideas to consider and the User Environment Design can capture and record large systems. But even when a company needs a large-scoped project, it is not necessarily the right thing to do. When companies are embarking on their first customer-centered projects, propose projects that are likely to be accepted, to give immediate value, and that will result in actionable changes. This will contribute to the grass-roots movement of introducing customer-centered design.

THE STEPS OF CONTEXTUAL DESIGN

The Contextual Interview—Getting the Right Data

To design a product that meets customers' real needs, designers must understand the customers and their work practice. Yet designers are not usually familiar with or experienced in the work they are supporting. If they operate from their gut feel, they rely on their own experience as a user. But designers generally are more tolerant of technology than average users, so they are not representative of end users.

On the other hand, requirements gathering is not simply a matter of asking people what they need in a system. A product is always part of a larger work practice. It is used in the context of other tools and manual processes. Product design is fundamentally about the redesign of work or life practice, given technological possibility. Work practice cannot be designed well if it is not understood in detail. Any introduction of change into a practice is as likely to make the practice less efficient as more efficient and delightful. So, requirements gathering means understanding how a practice is structured and imagining what technology might do to improve it—those improvements are what is required of the system.

You cannot simply ask people for design requirements, in part because they do not understand what technology is capable of, but more because they are not aware of what they really do. Because the things people do every day become habitual and unconscious, people are usually unable to articulate their work practice. People are conscious of general directions, such as identifying critical problems, and they can say what makes them angry at the system. However, they cannot provide the day-to-day detail about what they are doing to ground designers in what the practice entails and how it might be augmented with technology.

The challenge of getting design data is being able to get that level of detail about work that is unconscious and tacit. The first step of Contextual Design is Contextual Inquiry, the field data gathering technique that allows designers to go out into the field and talk with people about their work or lives while they are observing them. If designers watch people while they work, the people do not have to articulate their own work practice. If they do blow-by-blow retrospective accounts of things that happened in the recent past, people can stick with the details of a case using artifacts and re-enactment to remind them of what happened. Field data overcomes the difficulties of discovering tacit information.

In Contextual Design, the cross-functional design team conducts one-on-one field interviews with customers in their workplace focusing on the aspects of work that matter for the project scope. The Contextual Interview is based on four principles that guide how to run the interview:

- **Context**—Gather data in the workplace while people are working and focus on what they are doing.
- **Partnership**—Collaborate with customers to understand their work; let them lead the interview by doing their work. Do not come with planned questions.
- **Interpretation**—Determine the meaning of the customer's words and actions together by sharing your interpretations and letting them tune your meaning. When immersed in their real life and real work, people will not let you misconstrue their lives.
- **Focus**—Steer the conversation to meaningful topics by paying attention to what falls within project scope and ignoring things that are outside of it. Let the user know the focus so they can steer, too.

The Contextual Interview starts like a conventional interview, but after a brief overview of the work, it transitions to ongoing observation and discussion with the user about that part of the work that is relevant to the design focus. The interviewer watches the customer for overt actions, verbal clues, and body language. By sharing surprises and understandings with users in the moment, users and designers can enter into a conversation about what is happening, why, and the implications for any supporting system. As much as possible, the interviewer keeps the customer grounded in current work activity, but can also use work artifacts to trigger memories of recent activities.

The fundamental intent of the Contextual Interview is to help designers get design data: Low-level, detailed data about the structure of the practice and the use of technology within that practice.

Choosing Customers Depends on Project Scope. People are always asking how to shorten the Contextual Design process—time is always critical. Time in a Contextual Design project is directly proportional to the number of interviews conducted. The more interviews, the more time it takes to conduct and interpret them, and the more time to consolidate the data. So time is something that you can control as you choose how many people to interview.

But the number of people that should be interviewed is directly related to the project scope. The wider the scope, the more people need to be interviewed to cover that scope. A rule of thumb for a small scope, such as top 10 problems, usability improvement, next product release or checking a planned design, is 4–10 customers from 3 to 5 businesses covering 1 or 2 roles. Most large projects will not need more than 15–20 customers distributed over four businesses covering 3–5 roles. Very large, strategic projects might need 20–30 customers.

Choose the number of people to interview by balancing several variables, including the job roles to be supported, industries, and product use. When analyzing the project, first identify the job titles and the people that support the work. The work they do is what counts—if one person is called a system administrator and the other a database administrator, but they do the same work as it relates to the project, it does not matter that their job titles are different. The number of job roles needed to cover the design scope determines how many people must be interviewed. Interview a minimum of three people per job role; four is better.

If the product is to support people across industries, what matters is whether the industries are fundamentally different. Looking at real estate as an example, the work is structured differently within the industry: A group of small, distributed agencies; a large corporate real estate company; and an in-house real estate representative. In each situation, the communication, sharing, and work management are likely to differ, creating three different work practice patterns. So, to cover real estate, collect data in all three contexts. But if there are no changes to work patterns across industries—such as with the document editing component of an office tool suite, where editing is editing whether in banks, insurance companies, or software development organizations—then simply touch multiple industries without worrying about overlap. Try to interview at least two businesses per work pattern.

Geography works the same way. If culture and law affect the work practice, then to be global you need to gather data in multiple geographies. Try to interview two to four companies in a geographic area to ensure reasonable coverage. But if geography does not affect the practice, ensuring coverage is less important and data can simply touch on geographic differences.

Always try to look at customers that use homegrown and competing products, as well as the client's own products. Two to three instances of each of these are usually sufficient.

There is no need to make a grid and fill it all in. Instead, weigh and balance the variables ensuring a selection that has overlap in job title, industry, geography, and type of technology as appropriate to the project scope. The goal is to get enough repetition in the work so that each variable has 3–4 interviews that represent it. One person may represent three of the variables. As long as you have overlap, you will be able to find the common structure and key variations in the work practice. Remember, paper prototype interviews will expand the number of companies and roles represented in the whole project.

So, if time is a driver, cut back the numbers, but know that this will reduce the scope the project can cover.

Interpretation Sessions—Creating a Shared Understanding

Contextual Interviews produce large amounts of customer data, all of which must be shared among the core design team and with the larger, cross-functional team of user interface designers, engineers, documentation people, internal business users, and marketers. Traditional methods of sharing by presentations, in reports, or by e-mail do not allow people to truly process the information or bring their perspectives into a shared understanding. Contextual Design overcomes this by involving the team in interactive sessions to review, analyze, and manipulate the customer data. An interpretation team delivers the best results when its members represent a wide range of job functions and viewpoints. However, if the team is too large, the session will not be manageable. Usually 4-6 people are optimal, with 10 as the absolute maximum.

Although it is good to record interviews for backup, it is not advisable to transcribe the recordings or do videotape analysis. Both transcription and video analysis take too long, and video analysis also limits the perspective to one person. Instead, within 48 hours of each customer interview, gather the team in the design room, where the interviewer can tell the story of the interview from handwritten notes and memory.

Team members ask questions about the interview, drawing out the details of this retrospective account. At the same time, they perform one or more interpretation session roles: Recorder, model builder, moderator, or simple participant. The recorder types notes online, displayed on a monitor or LCD projection panel so that everyone can see them. These notes capture the conversation, including breakdowns and influences, in sequence. They are later transferred to 3M Post-it®Notes and used to build the Affinity Diagram.

Model builders hand sketch up to five work models onto flip charts (Fig. 49.1). Each work model depicts a different perspective of the work:

- The Flow Model depicts people's responsibilities and the communication and coordination required to do a job.
- The Cultural Model reveals influences on a person, whether external to the company (such as dependence on a vendor) or internal company policies.
- The Sequence Model shows each step required to perform a task in order.
- The Physical Model shows the physical layout of the work environment and the constraints it imposes on the design. It also shows the way people physically structure their work environment to support work.
- The Artifact Model shows how artifacts are structured and used, and suggests how their use could be extended.

Each team member brings a different perspective to the data, whereas open discussion enables the team to arrive at a shared understanding. The interactive nature of an interpretation session keeps everyone involved in the task at hand. Team members raise questions in the moment, triggering the interviewer's memory and eliciting more data than would be available from a designer working alone. Design ideas are generated and captured.

The interpretation session is a structured way to ensure that all relevant information from the interview is captured for use in the design process and shared with key team members. It is the context for both understanding and seeing the structure of the data and starting a real design conversation about how to address the user's needs with technology.

The work models structure what the team should look at, and look for, in the data. As such they ensure that the right data are captured. Team members learn that they need to draw these models so they become sensitive to capturing this data when they are in the field. In this way, the work models both provide a way to represent the data and a method to help people get the data they need for design.

Choosing Work Models Depends on Project Scope. The five work models and the affinity support coherent conversations focused on different aspects of the work. Each conversation will drive different design concepts, with some models forcing detailed thinking and others encouraging wide and broad design. The more models captured, the more work both within the interpretation session and during model consolidation. So, the number of models drawn affects resources and time. Depending on the project scope, different models are called for:

- **Identifying key issues:** Creating the affinity diagram will give you top issues, so to identify key issues only capture affinity notes.
- **Designing the next product release:** If you are looking to improve a product, but not reinvent it, you need only capture the affinity notes and the detailed models: The sequence, artifact, and physical layout of the work on the desktop or in the lab.
- **New product, market, or technology direction:** Detailed models only bring stepwise change. For the larger picture of the market and to push more innovative design, you also need to use the broader models: Flow, cultural, and the physical site model. These show you who the people are that could be supported and how they interrelate, what the value proposition is for this population, and how work is distributed across space.
- **Getting reusable corporate data:** Many companies have product lines that address the same work practice. Because the work models represent the fundamental structure of the practice, they tend to be stable over time. The wider data, once collected, can be expanded and reused by the same team and by teams producing related products. Collecting the wide-scoped data at some point and building upon it is efficient in the long run.

What models you capture affect the number of people needed in the interpretation session. To have a smooth interpretation session and capture all the models, you must have an interpretation team of four. A two-person interpretation team is the minimum. At least one outside perspective is needed: One other person to question and prod the interviewer. A three-person team ensures reasonable diversity and group sharing, even when fewer models are used. However, if you only have a two-person

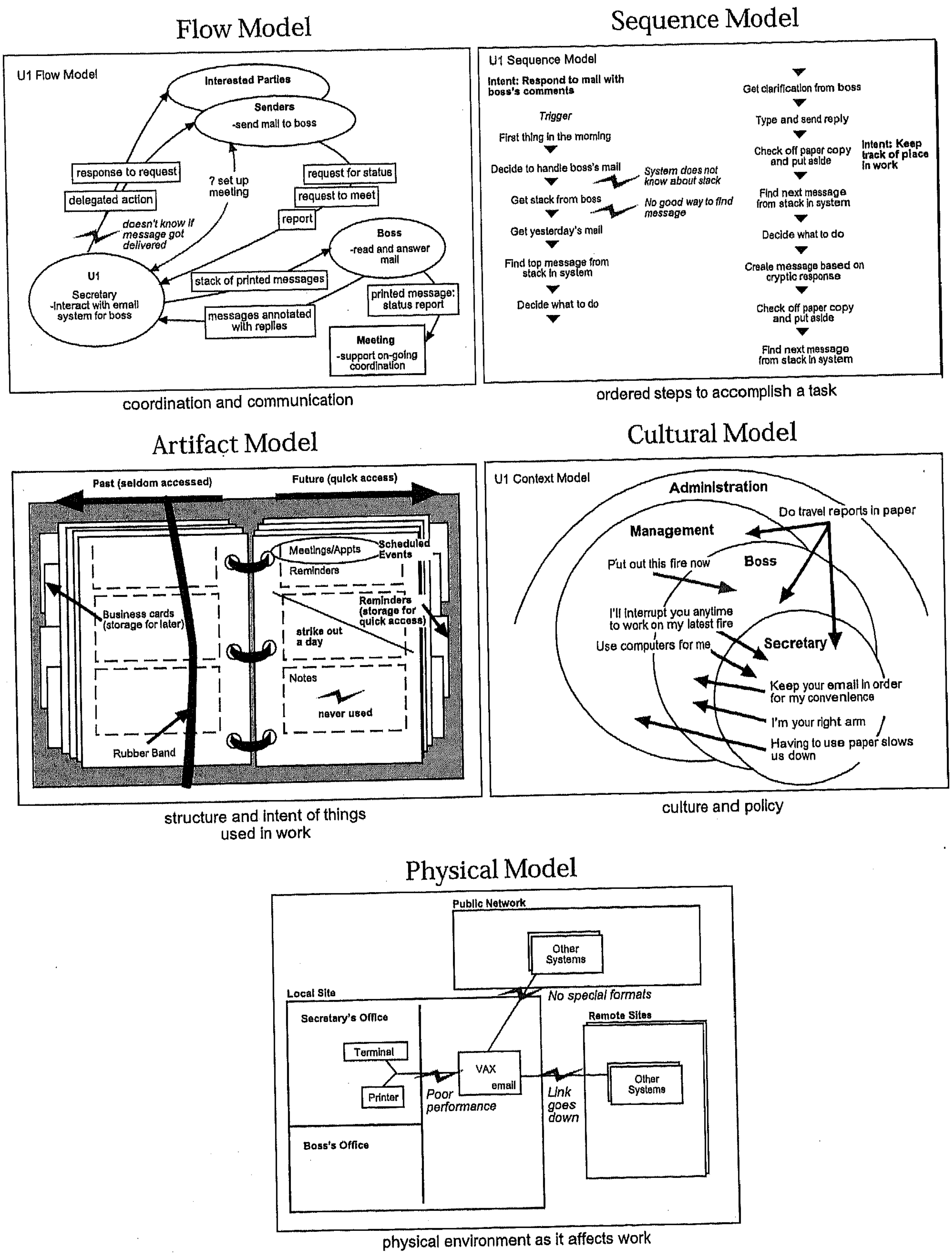

FIGURE 49.1. The interpretation session results in affinity notes and five work models.

core team, you can extend your team by rotating in extra people. In this way, you can increase buy-in with limited involvement (2 hours) and benefit from outside perspectives without requiring that these adjunct team members leave their regular jobs for long periods of time.

Consolidation—Creating One Picture of the Customer

Consolidating the work models enables teams to create one representation of a market so that the design addresses a whole population, not just one individual. The most fundamental goal of Contextual Design is to get the team to design from data instead of from the "I." If you walk in the hall and listen to designers talking, you will hear: "I like this feature." "I think the dialogues work best this way." "This user interface component doesn't make sense to me." It is rare to hear, "Our user data says that the work is structured like this, so we need this function." It is natural for people who design to make a system hang together in a way that makes sense to them. But they are not the users, and they are not in any way doing the work of the people they are designing for. Getting designers, marketers, and business analysts out to the field and into interpretation sessions moves them away from design from personal preference, but you do not want them to become attached to their user to the exclusion of the rest.

Product and system design must address a whole market or user population. It must take into consideration the issues of the population as a whole, the structure of work, and the variations natural to that work. The core intent of building consolidated models is to find the issues and the work structure and create a coherent way to see it and talk about it. Because the work models and affinity diagram separate and focus conversation on different aspects of work, consolidation creates six points of view from which to see and discuss the data.

Because consolidated models segment work practice into coherent and meaningful chunks, the team is able to see, capture, and think about very complex and rich information. Complexity often leads to overwhelm, and overwhelm leads to a desire to simplify or reduce data. Data reduction, in the form of identifying the top 10 findings or creating a simple summary, loses the complexity and structure of the work that the interviews revealed. Consolidating the data and systematically walking and discussing each model as a team helps the team think about complex information without getting overwhelmed.

While discussing the key factors of each model, team members will naturally synthesize the findings into a reasonable view of what a product needs to support. Given this data, any good designer will be stimulated to imagine how their product can better support the work. So systematic interaction with consolidated models becomes the basis for design thinking and informal prioritization of customer needs.

The Consolidated Models. Each model focuses the team on a different set of issues to consider.

The affinity diagram (Fig. 49.2) brings issues and insights across all customers together into a wall-sized, hierarchical diagram. Using the Post-it®Notes created in the interpretation session, team members organize all of the data one piece at a time, finding common underlying themes that cross the customer population. The process exposes and makes concrete common issues, distinctions, work patterns, and needs without losing individual variation. Walking the affinity diagram allows designers to respond with design ideas that extend and transform the work. They write these on Post-it®Notes and stick them right to the data that stimulated the idea. This encourages a culture of design from data over a focus on "cool ideas" generated from the "I."

Building the affinity after a good cross-section of users (usually 6–12) has been interviewed enables the rest of the interviews to be increasingly targeted to address specific issues that may emerge. The affinity is built from the bottom up, which allows the individual notes that come together to suggest groupings instead of trying to force them into predefined categories. Groups are labeled using the voice of the customer—saying what they do and how they think.

The consolidated flow model (Fig. 49.3) identifies the key roles played by individuals and combines them across individuals to show the roles that a system must support. This is your war map, representing the players in the market to target and support. The flow model helps the team see the pattern of relationship between players and reveals opportunities for collaboration; it shows how responsibilities are clustered into roles that can be supported coherently or automated; and it reveals artifacts and information that pass between people that might be supported online.

The consolidated cultural model (Fig. 49.4) is important to marketing because it reveals the value proposition. It collects and shows the influences, constraints, interpersonal friction, policies, standards, and law that people work under. It reveals positive feelings and irritations, including those with the vendor collecting the data. As such, dialogue with the cultural model reveals primary values and irritations in the market being addressed. A product that supports those values or removes those irritations has a good value proposition story. Because cultural differences, policy, and law are also revealed, it shows the design team what they must consider within the design for adoption. Think of it as the "get real" model, because it makes culture and the need to design for the culture real to the team.

Consolidated physical models (Fig. 49.5) reveal the common aspects and variations of physical structure across a customer population. Like the cultural model, the consolidated physical model is a get real model. The physical model reveals the physical constraints of distance, time zone, and overall physical and system access. It reveals the way space is used and how people set up their offices or workspaces to support their work. People's piles are a window into their mind and how they organize the work. The physical model, like the affinity, flow, and cultural models, is a big picture model; it pushes open design thinking and shows the whole social, physical, and cultural environment that the team will have to support. But it is a swing model because it is also a detailed model, revealing how people separate and organize the details of the work. It will guide detailed design and storyboarding, as will the consolidated sequence and artifact models.

The consolidated sequence model (Fig. 49.6) shows the detailed work structure that the system will support or replace.

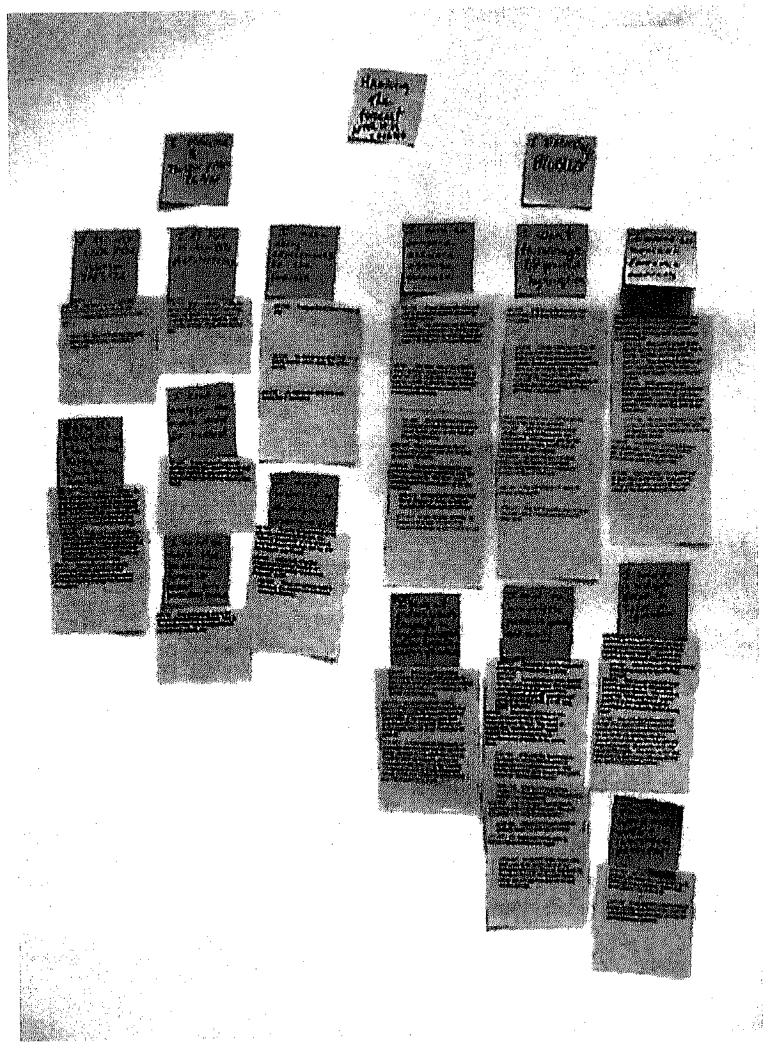

FIGURE 49.2. The affinity diagram.

This is similar to traditional task analysis, showing each step, triggers for the steps, different strategies for achieving each intent, and breakdowns in the ongoing work. Work redesign is ultimately about redesigning the steps in the sequences. Whether the redesign eliminates or changes the steps or eliminates the whole sequence, knowing the steps and intents keeps the team honest. More often than not, technology introduces new problems into the work by failing to consider the fundamental intents that people are trying to achieve. Redesign will better support the work if it accounts for each intent, trigger, and step. This does not mean leaving the work the same; it means that the team has seen the current activity and has completely considered what will happen to it in the new work redesign. The consolidated sequence is critical as a guide for storyboarding.

Consolidated artifact models (Fig. 49.7) reveal the structure of the artifact, how it is used, how it is presented, and the data that it represents. Consolidated artifact models help the team redesign artifacts that have been identified as candidates to be put online. Sometimes teams want to eliminate artifacts they think serve no useful purpose, so consolidated artifacts also keep the team honest. Any time a team wants to eliminate an artifact, they must ensure that all the intents of that artifact are fulfilled in some other way by the design.

How Can So Little Data Characterize a Whole Market? Consolidated models create data that is reusable for years. For example, we have been collecting data on system management for 13 years across multiple companies, and the basic flow model roles and high-level activities and intents are the same. The core of any work practice does not change much over time. Technology usually introduces changes at the level of the step in the sequence model. Aspects of the roles may be automated, but the role itself remains, perhaps now played by automation instead of a person. The values that drive the practice, the benefits, persist. Once they are met, they become latent, expected needs instead of unfulfilled needs.

Lasting consolidated data is built anywhere from 15 to 30 field interviews and can characterize markets of millions of

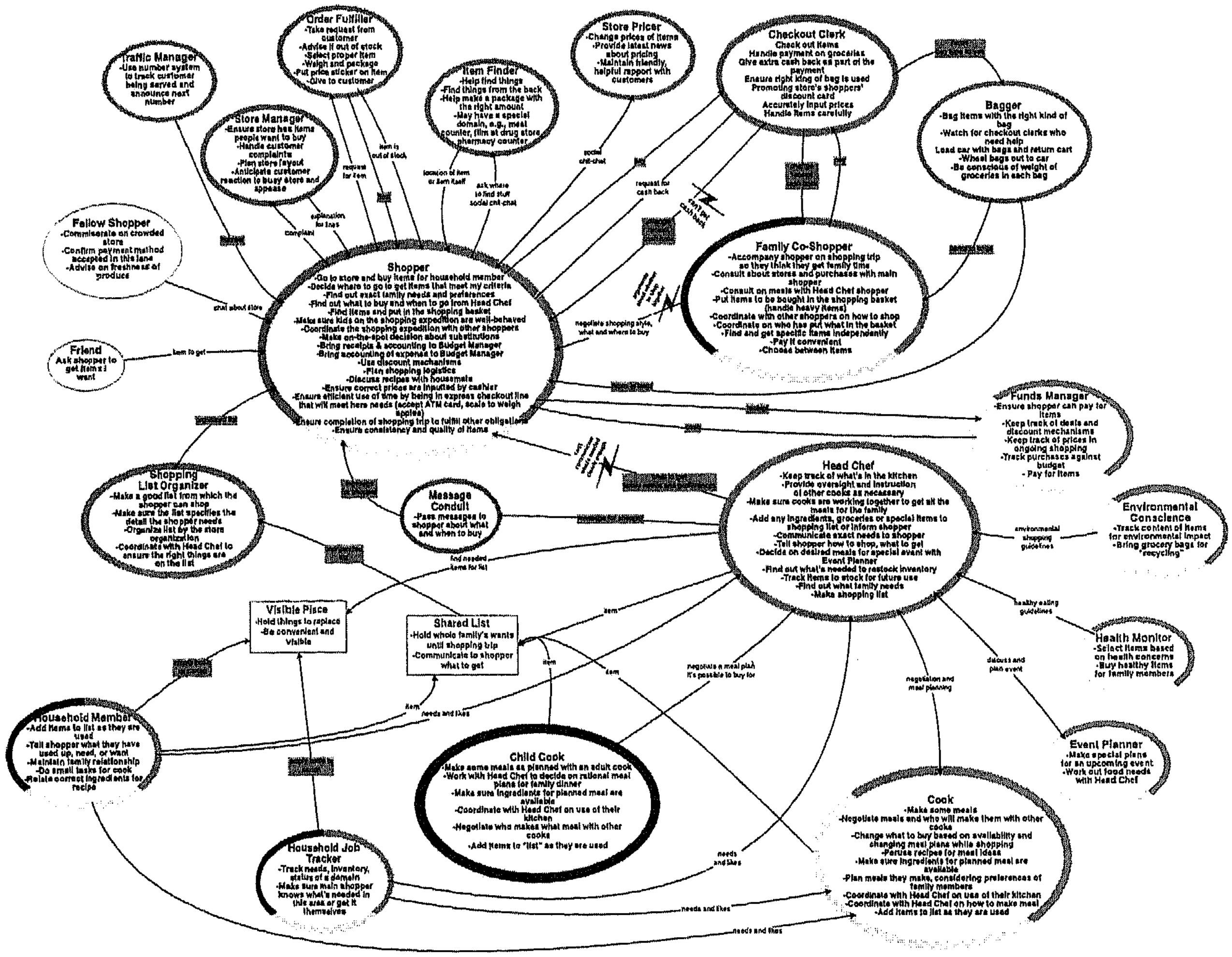

FIGURE 49.3. The consolidated flow model.

people. Even consolidated data from 8 to10 people can identify a large percentage of the key issues that will eventually be identified in the market. For example, if you consolidate after 10 interviews, you can grow the models and affinity from there. An early affinity can show holes in the data to guide further data collection, but already represents the key areas and distinctions that will grow in detail as more data is added. Similarly, additional data adds depth and detail to all models, but the basic structure, central to the project focus, is identifiable early on. This means that a little bit of data goes a long way. If this data is also stable over time, it becomes a key corporate resource.

Early on in usability testing, people were not comfortable with test results from small numbers of users. Whatever the formal arguments, 15 years later the industry has learned empirically that small numbers are enough. Contextual Design is engaged in market characterization at the level of work practice, so it is compared with marketing studies with random samples and hundreds of data points or to focus groups with 10 people in each session. How can so little data characterize the practice of so many people? Moreover, how can this data be stable for years?

Contextual data does not claim to show trends. Contextual data cannot be counted because it is based on observing people in the field and each person's activity directs what is seen. You simply do not get the same data from each person. Under these conditions, frequency measures and trends have no meaning. Contextual data shows the structure of the practice. Contextual Design gets its power from designing from an understanding of work practice *structure* without losing variation.

The reason that people get hung up on numbers is that we tend as human beings to focus on our variation, our differences, rather than our basic similarities. A focus on variation leads us to design products with near-infinite customizations. These either destructure a system with ever more customizable and slightly different options or require costly upfront customizations at installation. But, if people are all so different, how could markets exist? Clearly, there is enough similarity of practice between people for the idea of shrinkwrap software and off-the-shelf electronic products to make sense.

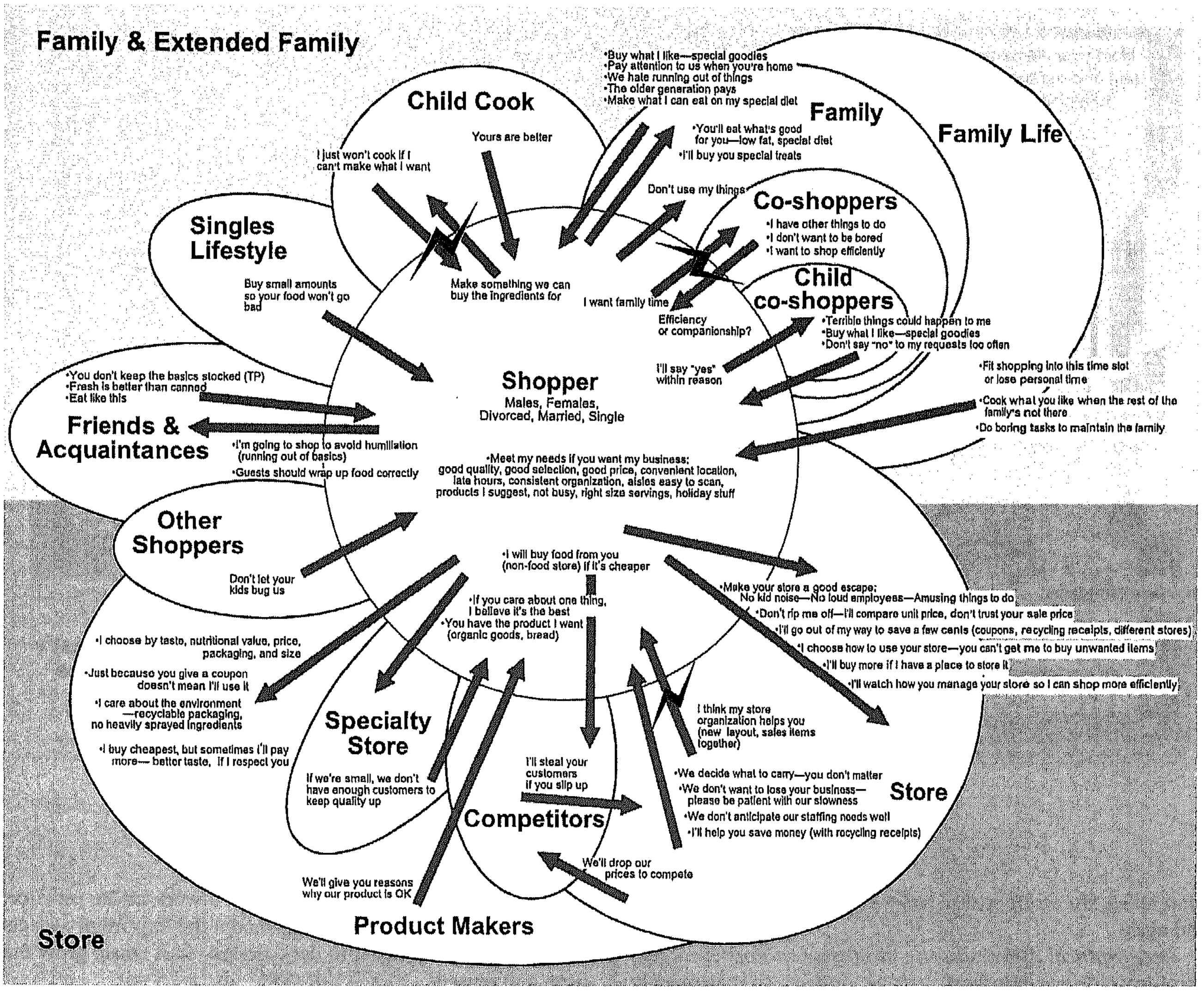

FIGURE 49.4. The consolidated cultural model.

Consolidation helps people see and find the structure in work practice, and this drives successful design. Here is a way to think about it.

We're all different, but we're all alike: Everybody looks different; humans have great variation. People are of different ethnic groups, cultures, and child-rearing practices. Everyone chooses different clothes, hobbies, careers, and lifestyles. So, at one level, we are all unique. But at the same time, we are all alike; we are human beings with one head, two arms, two legs, and so on. From the point of view of clothing manufacturers, we all buy clothes off the racks in department stores; tuning a few preferences in the hemline is enough. Structurally, the variation between people is small and the structure of our bodies is common. So, although we tend to focus on our differences, given the project focus of clothing manufacturing, we are pretty much the same.

There are only so many ways to do work: Any product and system design is really a very narrow focus on the human experience. Within one kind of work, there is only so much variation possible. The roles we play, the intents and goals we have, and the way we do things is common. Variation, once you start looking for structural elements like roles instead of job titles, is small. Experience shows that there are only two to four strategies for any primary task in a work practice. If those are the key strategies for that work practice, the question is not which we support, but how we support them all. Frequency data is not needed to show that these strategies matter; these are the basic elements of the practice. Designs that target a certain domain of work are created for a certain set of people doing a fixed set of work tasks, situated in the larger work culture, using the same set of tools, trying to achieve the same kind of goals. Under these constrained conditions, the work practice will have similar patterns. If this were not the case, there would be custom systems for every single person. Instead, people use the same software systems with some preferences and options. After you have collected data from three to six people doing the

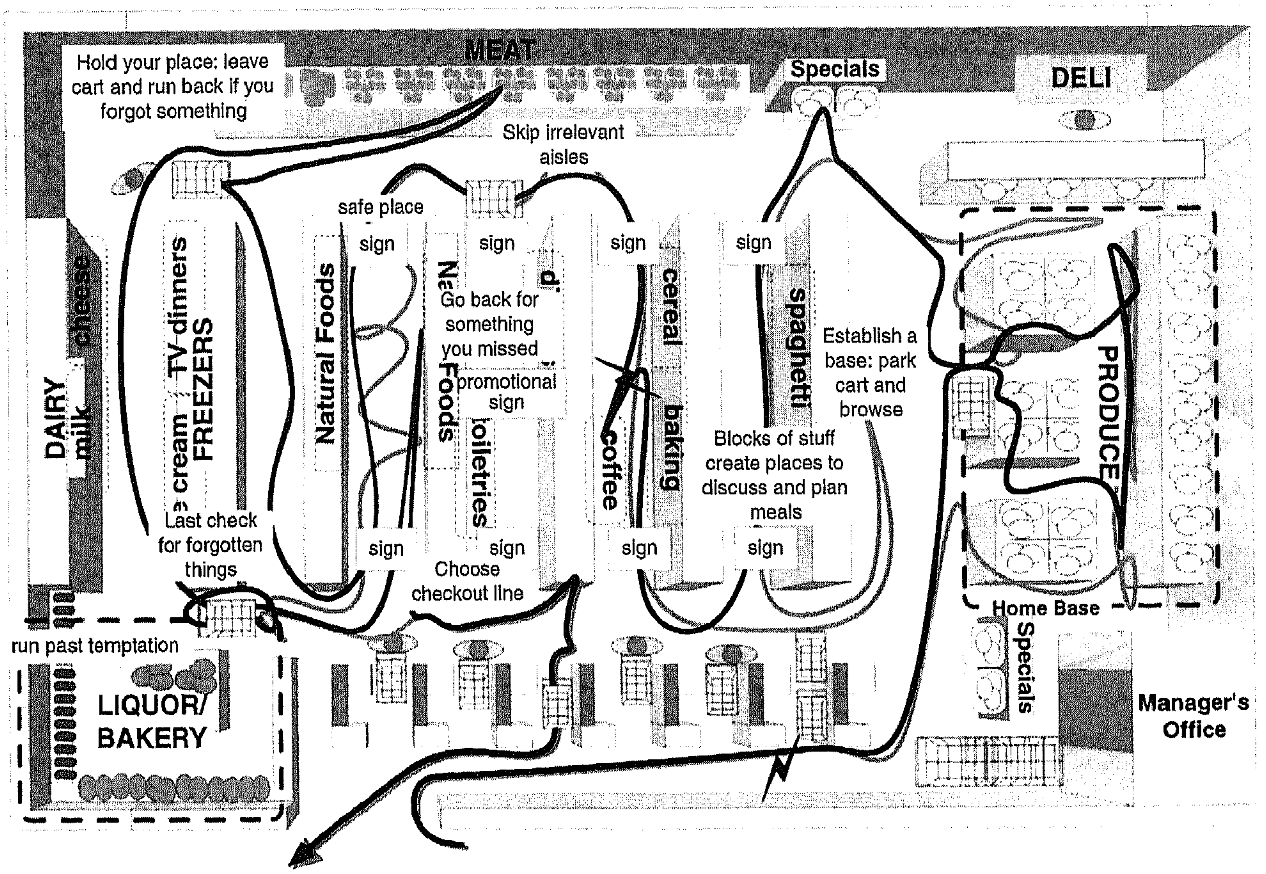

FIGURE 49.5. The consolidated physical model.

same thing, the work pattern and issues start overlapping again and again.

Geographical differences are mostly about law: Even geographical differences in work practice are small. They are mainly about law or lawlike standards, not culture. For example, in Europe, if somebody travels for their job, they are compensated using per diem rates. The actual rates and inclusions vary from country to country, but they are always paid per diem. In contrast, in the United States, they may be compensated per diem or they may need to show their receipts. In any event, because there are only two ways to compensate work travel, there are only two possible interfaces—a receipts-driven and a per diem interface. Cultures are interesting because food, values, personalities, and lifestyles are different. But, from the narrow view of work practice, the variation that affects practice the most tends to be legislated by corporations or government. Other differences between cultures are best understood as variations in emphasis; some cultures emphasize one strategy or cultural value more than another, but most cultures have people in them that represent all the different values and strategies anyway. So, it is possible to sample across cultures and check at the extent of difference before deciding how much global data is needed to characterize the market.

With infinite time and resources, we could satisfy our worry about complete coverage by extensive customer sampling. Luckily, because a design is targeted at a work or life practice, a small amount of data will let us model the important aspects of the practice and use it to drive design. Start small, grow the data, and reuse it.

The number of interviews affects the timeline of a project. What you consolidate also affects the timeline of a project. You do not need to consolidate models that are not going to help the project, even if you captured them. For example, if you are not redesigning artifacts and the artifacts are not relevant to the design, consolidating the artifacts is a waste of time. You can build the physical model to help the team understand what happened in the interview, but if the arrangement of the details of the model does not impact the problem you are solving, then there is no need to consolidate the physical models. On the other hand, do not postpone consolidation because you have no time now. It will not be possible to recover the context of the interview sufficiently to consolidate data later.

Visioning a New Work Practice

Visioning is about invention. The question is, invention of what? One might argue that the first form of technology was the stick. The stick is an extension of the arm and finger. A good stick is narrow enough to fit in the space that the arm or finger cannot fit.

-Activity 2. -Stop and Buy

1) Trigger: Stop If: a. Passing a block of stuff needed, or staple, or treat (may or may not be on list)	▪ Get ingredients for recipes ▪ Please co-shopper (especially kids) by buying them a treat
b. Reached a "home base" where you're willing to browse, park cart and wander	▪ **Many shoppers scan items, not signs; signs are invisible to them even if they advertise specials they care about** ▪ Allow store to suggest purchases after I've decided I'm in an area of the store where I'll allow that ▪ Save pushing the cart around by parking it and using it as a home base while I wander from item to item
c. You want to check on prices for future reference (may or may not buy)	▪ Save money on staple items
d. You notice sale item and want to decide if it meets need	▪ Save money on frequently used item, even if not on list
2) Use block of stuff to suggest possible meals a. Discuss menus with other shoppers	▪ Plan the week's meals taking into account what is in the store and what looks good ▪ Share the meal planning ▪ Plan who will cook what
3) Decide whether to get stuff from that block, considering: a. What exact type and brand to buy, what family will like, size, weight and amount to buy, cost, quality, ripeness, freshness, good packaging, whether I have coupons, and ingredients	▪ Choose a food that supports my values ▪ Compare and learn quality of new items ▪ Minimize sugar/fat consumption
4) Put items in basket	
5) Or, discover store doesn't carry desired item	
6) When done with the block of stuff, check to see where you are in the aisle a. See something, remember need – Triggers other items you need b. Backtrack if missed something on the other side of aisle, or c. If you're reminded of something you forgot 1) Go get the item (*Beeline* shopping strategy) 2) Go back to the cart and make sure everyone's OK 3) Put item in basket	▪ Reorient yourself after buying from a block of stuff ▪ If blocks on opposite side of aisle are staggered, may have to backtrack ▪ Get items quickly

FIGURE 49.6. The consolidated sequence model (partial).

A good stick is long enough to reach what the arm cannot. Technology will always need to fit within a larger human practice. So, design is effectively the invention of practice in which technology fits well. Design of technology is first design of the story showing how manual practices, human interactions, and other tools come together with your product to better support the whole practice. Visioning is the Contextual Design technique to help teams tell that story. Visioning is a vehicle to identify needed function in the context of the larger work practice. Visioning ensures that teams postpone lower level decisions about implementation, platform, and user interface until they have a clear picture of how their solution will fit into the whole of the practice. Teams commonly focus too much on low-level details instead of the full sociotechnical system. This is one cause of breaking the work and failing to create something the market wants. So, the primary intent of visioning is to redesign the work practice, not to design a user interface. Because a visioning session is a group activity, it fosters a shared understanding among team members and helps them use their different points of view to push creativity.

The first step to a visioning session is to "walk the data" model by model, immersing the team in customer data so their inventions will be grounded in the users' work. During the walk, team members compare ideas and begin to get a shared idea of how to respond to the data. Nobody should get to vision unless they have participated in walking the data. Without this rule,

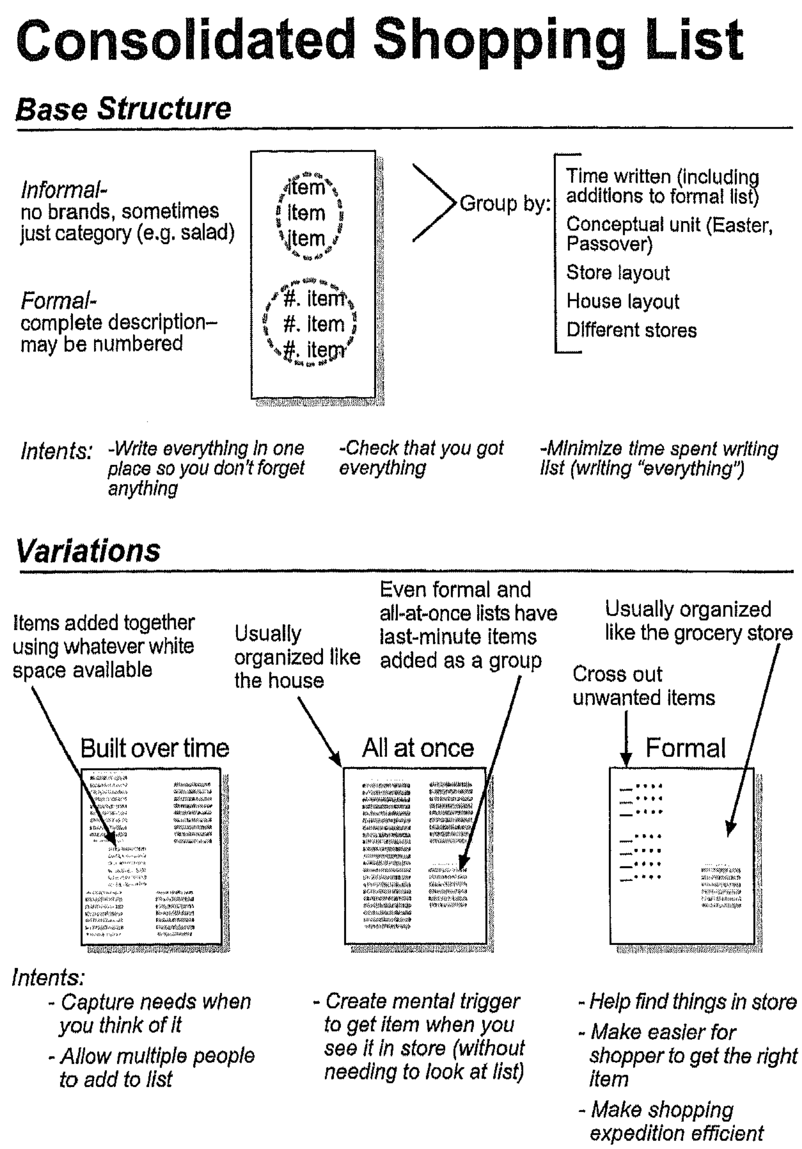

FIGURE 49.7. The consolidated artifact model.

the process is no longer data driven; anyone can walk in and offer their pet design ideas. Simply walking the customer data naturally selects and tailors pre-existing ideas to fit the needs of the population. Because visions are evaluated, in part, on their fit to the data, knowing the data is important.

During the visioning session, the team will pick a starting point and build a story of the new work practice. One person is assigned to be the "pen" who draws the story on a flip chart, fitting ideas called out by the team members into the story as it unfolds. The story describes the new work practice, showing people, roles, systems, and anything else the vision requires. The team does not worry about practicality at this point; all ideas are included. Creating several visions allows the team to consider alternative solutions.

After a set of visions is created, the team evaluates each vision in turn, listing both the positive and negative points of the vision from the point of view of customer value, engineering effort, technical possibility, and corporate value. The negative points are not thrown out, but used to stimulate creative design ideas to overcome objections. When complete, the best parts of each vision and the solutions to objections are brought together into one, synthesized work practice solution (Fig. 49.8).

How Can You Invent From Customer Data?. Every team raises this question. Customer data tell you what is, not what could be. How can you see the future by looking at the past?

Every invention supports a real need, else why would anyone want it?: Invention is a response to some life or work practice by a designer or technologist who, seeing a need *and* knowing the technology, imagines a new possibility. Edison did not invent the idea of light; he saw candles and gas and invented light bulbs. Dan Bricklin (Beyer, 1994) did not invent spreadsheets;

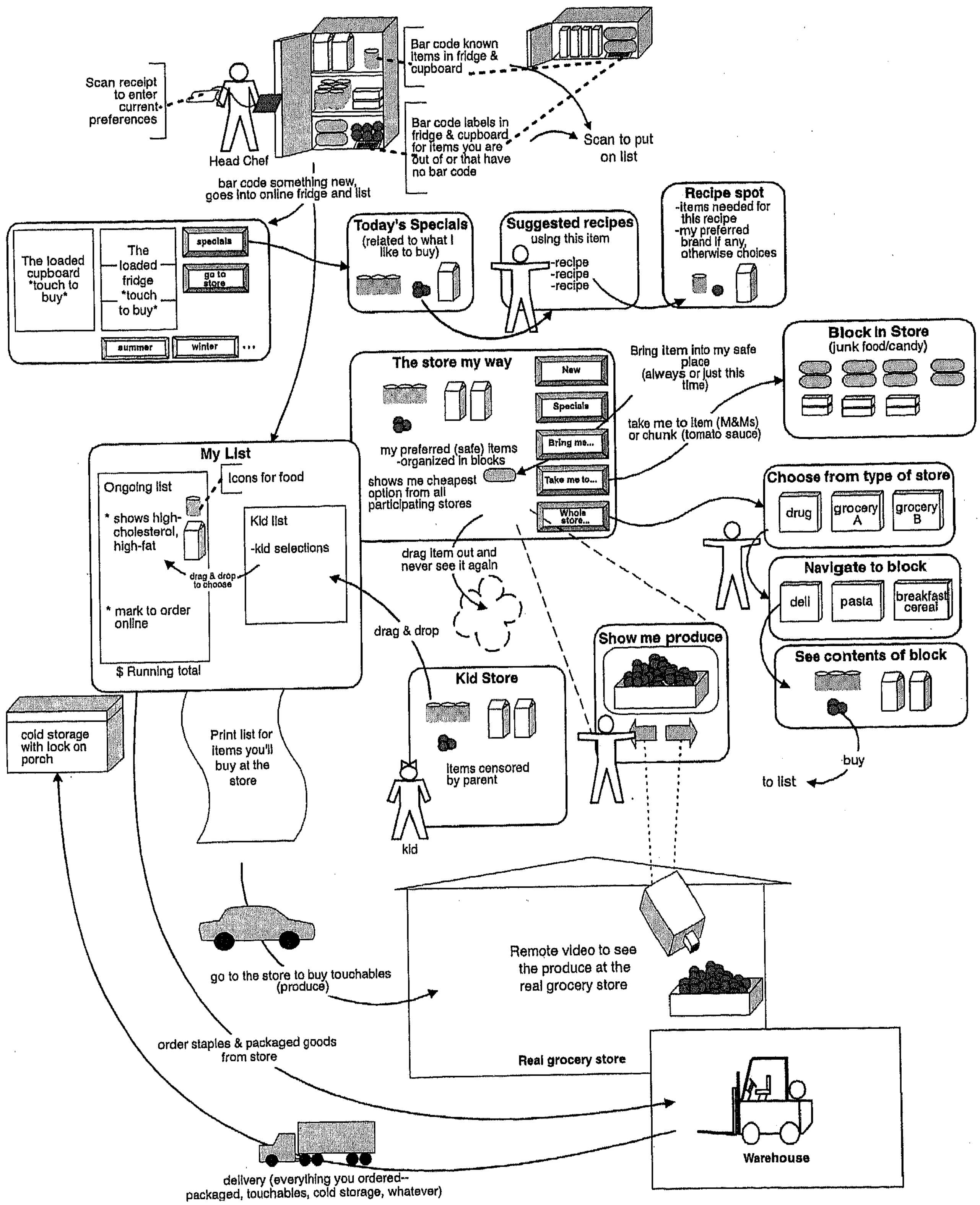

FIGURE 49.8. A complete, synthesized vision.

he saw how paper spreadsheets are used and knew what technology could do. The developers of WordPerfect worked in the basement below a secretarial pool. Nobody invents entirely new things that fulfill no need and contribute in no way to human practice; they invent new ways to fill existing needs and overcome existing limitations. As people incorporate these products and new ways of working into their lives, they reinvent it by adopting and adapting the new way of working. Thus, if designers are out there seeing people living their lives with, without, and in spite of technology, they can see future directions for technology.

A mobile phone businessman tells a great story when asked about how you could have invented the hand-held cell phone by looking at customer data: You would be driving down the road and watching people suddenly pull off the road and park their cars to make a phone call. You would see them pulling into gas stations to make a telephone call. They would be out in their yard in the midst of yard work and run inside to make a telephone call. You would see people looking all over the office and calling over the loud speaker to find a person not at their desk. You would see families call all their child's friends looking for them. Then, knowing phone technology, you would ask yourself what if, when people called, the call went right to them? Gee, how could we, the makers of phones, do that.... So, where do you think the idea of cell phones came from, anyway?

A vision is only a good as the team's combined skill: Customer data is the context that stimulates the direction of invention. But there is no invention without understanding the materials of invention: Technology, design, and work practice. The visioning team needs to include people who understand the possibilities and constraints of the technology. If the team is supposed to design web pages and none of them has ever designed a web page, they will not be able to use web technology to design. When the people who always designed mainframes were told to design WYSIWYG interfaces, they replicated the mainframe interface in the WYSIWYG interface. Similarly, if the team has no interaction designers or work practice designers on it, the resulting vision will not be as powerful as if they did. This is why design teams should include people with diverse backgrounds representing all the materials of design and the different functions of the organization. Only then will an innovative design, right for the business to ship, emerge.

Even new technologies can be guided by customer data: Even if you are looking for ways to capitalize on an entirely new technology, there is work practice to observe and gather data about. A designer might say, "We're creating something that has never existed before. There's no customer data to collect." However, there was a time when computers did not have sound and no one knew what it would do. Was there no data to collect to guide this technology? Was this *really* brand new? Sound was once a totally new *technology*, but *sound* is not new. Using sound appropriately in computers is a challenge. What data should you collect? Data on the experience of sound. Walk through the world seeing what people do with sound. Listen, look, and talk with them about random sound, talk, music, sound in the environment, and the role of sound in people's lives. Also pay attention to silence, because you must not violate that. Studying sound and silence, noise and communication, reveals opportunities and identifies appropriate ways to use sound on a computer.

Don't ask customers what to make; understand what they do: Customers who are not aware of the details of their own practice, who do not know the latest technologies, and who do not know what your business is capable of cannot tell you what to invent. Do not ask them. Instead, understand what they are doing and capture it systematically. Immerse a design team—that *does* understand technology, work practice, and the business—in that data and let them vision. But the vision has to be right for people, so take it out and test it. Let people test drive the future in our paper mock-ups and let their tacit knowledge of their lives shape and direct the vision.

Storyboarding—Working Out the Details

Storyboarding provides the opportunity to work out the detail of the vision. Too often when people design, they break work practice because they jump from their big idea to low-level user interface and implementation design. As soon as designers start focusing on technology, technology and its problems become their central design concern. How technology supports work or life practice is subordinated. This also happens when people design from idealized models of how processes work. A classic story is about an ordering system that broke the work of the corporation. No one knew that the person answering the phone was a good friend of the person prioritizing the orders to be fulfilled. They had created an informal "hot order" channel that was never modeled in the new, planned process.

The steps and strategies of a work practice, being tacit, are easy to overlook. But if you are out in the field watching them, you can see them. This is what is captured in the sequence models. Storyboarding keeps the team honest and the design clean. Guided by the detailed models—the sequence, artifact, and organization of the work place (physical)—the vision is made real in storyboards. Storyboards ensure that the team does not overlook any intents and steps that are critical to the work. Even if the work is changed, think through the details of how it will be changed to ensure that adoption is easy.

Storyboards are like freeze frame movies of the new work practice (Fig. 49.9). Like storyboarding in film, the team draws step-by-step pictures of how people will work in their new world. Storyboards include manual steps, rough user interface components, system activity and automation, and even documentation use. Storyboards are similar in structure to the consolidated sequence model and incorporate ideas from the other detailed models. They show what happens when the roles on the flow interact with the system and each other. Like consolidated sequences, they show how the system will handle multiple strategies.

Because they focus on work practice design, storyboards prevent the design team from prematurely delving into too much detail. They are guided by data, and after each task has been thought through and sketched, the team reviews it to ensure that it remains true to the customer data. This does not mean that no invention happens, but the team must account for the steps and other data elements of the models. They must look at them

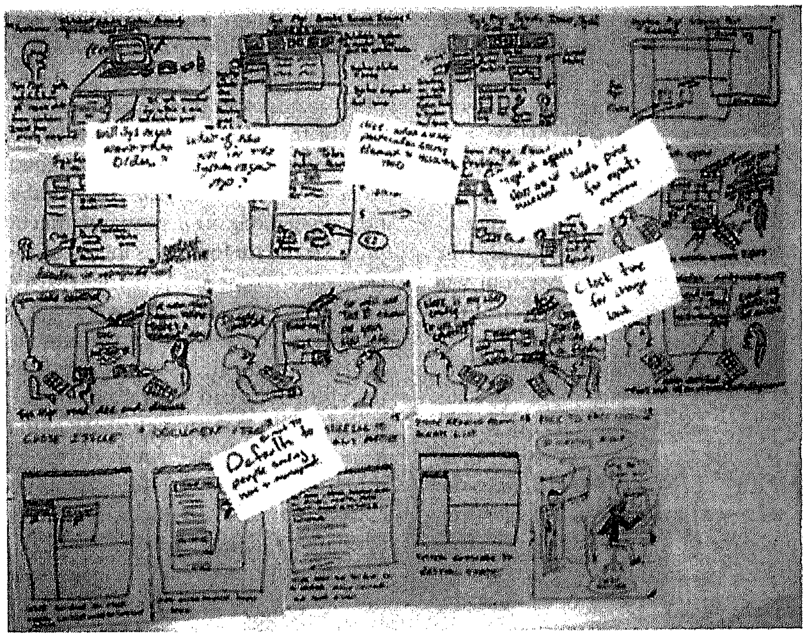

FIGURE 49.9. Storyboards are like freeze frame movies.

and make a conscious decision about how to handle them. They might change all the steps and even eliminate whole sequences; as long as people can still achieve their fundamental intents, the change will work. When teams forget or ignore the user's intent, the design is in trouble.

The vision is a natural "prioritizer." It is not necessary to storyboard every sequence. Instead, identify which sequences are relevant to the vision and storyboard those. The sequence models should represent the core tasks to achieve the work within project scope. Supporting those sequences will be enough for a next product release.

How Is Storyboarding Different From Building Scenarios? The storyboard is fundamentally a future scenario guided by the vision and reined in by the data. Every system implies multiple storyboards of use. Just as a kitchen supports multiple scenarios of cooking (the quick hot dog, the Thanksgiving dinner, the breakfast for the toddler), a system supports multiple scenarios of use. A storyboard is a high-level use case.

Any vision implies multiple scenarios to fulfill the vision. No system should be built around one storyboard or one future scenario. Gathering a set of people together in a room to "invent the future" with future scenarios, returns them to design from the "I." Where do the cases come from that the system must handle? If not from data, they must be generated by the people in the room. If they come from idealized process flows, they are probably unrealistically linear, ignoring the multiple strategies and complexities of real work processes. Good systems are not linear; they do not force or follow a single linear path. Driving scenarios from consolidated data ensures that different strategies for doing a task will be considered and that the design will support the different routes that real people use to achieve their intent and the different roles that people play to get their work done.

Storyboarding consolidated sequences ensures that you maintain the breadth of real work without having to make up cases. Thus, if the team is missing sequence data that supports the vision, they should collect it. But now, knowing what data is needed, data collection can be very focused. Get sequence data from two additional interviews and consolidate just those two sequences. If you can only get one sequence, analyze it as you would for sequence consolidation. Having a guide to the design is always better than making it up. Teams that try to design the future without data get lost, confused, and are prone to overdesign. Data keep teams focused and centered so invention is reasonable and likely to succeed.

User Environment Design

A product really has three layers of design. The user interface is the access mechanism to the function, structure, and flow of the work the user needs to do. The implementation (object model) is the way that the system will make that function, structure, and flow happen. But the core of a product is that middle layer: The explicit work the product is performing. The problem is that the work layer feels invisible. People know what a user interface is and they know what an object model is. But where is the structure of the work of a product? An architect has such a representation for a house design: The floor plan. It shows structure (the rooms each with a purpose and its function), function (what you can do in each room), and flow (the hallways and movement between and within rooms). Having a floor plan gives the architect a way to represent how people will live in the house without worrying about the interior decoration or the wiring system. The User Environment Design is a software floor plan; it represents the structure, function, and flow of a system.

The User Environment Design (Fig. 49.10) separates conversations about the system's underlying structure from conversations about its user interface and implementation. Like the architect's floor plan, it focuses the team on designing the work structure and flow within the system. To support work practice, designers must ensure that the product fits the larger work practice represented by the vision and storyboards, and that the work hangs together within the system itself. The User Environment Design enables the team to view the whole system and the relationships between its parts. It shows how the system will structure the user's work and how it will interface with other systems. It forces the team to focus on the system's function instead of jumping ahead to user interface issues or beneath to implementation issues. This helps ensure a workable and delightful user experience because it keeps the work coherent.

The User Environment Design formalism provides *focus areas* for each activity. Each focus area is a place that defines the functions and objects to be accessed from that place. Links from one focus area to another define the flow within the system that the user can traverse. Because focus areas support each other and naturally cluster related function and related focus areas, the User Environment Design ensures that designers think about the work pattern within the system without being distracted by discussions of icons, placement of function, or words on the user interface.

The User Environment Design was formalized before people were using site maps to guide Web site design. But both web-based and traditional products and systems benefit by having an abstract representation of the whole system that designers can look at, talk about, plan from, and test against scenarios of use. It helps designers and marketers get beyond thinking of the product as a list of functions or features. Because it shows all the parts of the system, it becomes a tool for system planning.

Managers can use the User Environment Design to see how their product interfaces with other products. The User Environment Design helps ensure design from a systems perspective. They can create rollout plans that carve up the system into coherent releases, each of which contributes significant benefit to users. They can assign parts of the system for further design to individual developers who can now see how their part fits into the whole.

Just as consolidated models help you see the structure of the work without getting lost in the variation, the User Environment Design helps teams see the structure of the product without getting lost in its details. Maintaining work coherence within and outside of the system requires this ability to conceptualize structure.

The User Environment Design Works for New Systems: What Do We Do for Existing Systems? Most people can see how building a User Environment Design makes sense for new systems. But most of the time, teams are modifying existing systems, not designing new ones. Existing systems are overwhelming. They are full of legacy features, and they demand that existing users continue to be supported. You cannot just start over.

Implicit in every kind of software is a User Environment Design, just as every house has a floor plan. You just have to find it. You can redefine and extend an existing system by first building a Reverse User Environment Design to show how that system is structured. Once people see their product structurally, they often realize why people are having a hard time with it. Is your system too leggy? Does it have a star structure requiring the user to go back and forth to a central place to get anywhere? Does your system mirror your database hierarchy? Is the function clustered in your system so that every focus area has a clean purpose, or are functions mixed so the purpose of each place is confusing? These are questions that a Reverse User Environment Design can answer.

With a Reverse User Environment Design, you can plan to watch for these structural problems in the field interview. After the vision, you can determine whether to use the Reverse User Environment Design as the base of the new system, building on it and changing it in response to the storyboards. Used this way, it will constrain the future design and ensure that the design team deals with version-to-version compatibility. Or you can open up the design and create a new User Environment Design, opportunistically taking existing focus areas and functions from the existing system to include in a new design.

Good design is about seeing structure. The User Environment Design helps the design team think about and design for structure in their own and competitive products. Once teams can see structure, they can design for it, whether it is a new product or an existing one.

User Interface Design and Mock-up

At some level, the User Environment Design is a theory. If you were right about how you characterized the market and if you picked the right issues to address and if your vision of the new work practice was a good one and if you translated that vision through the storyboards and into the User Environment Design well, then the system should support and extend people's work. How can you find out?

Users do not understand models. Pelle En (1988) understood this years ago when he realized that holding users hostage in a conference room, squeezing object modeling insight out of their tacit understanding of their work, was painful and inaccurate. Researchers at Aarhus (Kyng, 1988) started using mock-ups of new design artifacts to test their ideas by having people use them in real work situations. They found that, within that context, users could understand the technology and further define the requirements. Users can talk user interface talk, or form factor talk for physical systems, but cannot talk model talk. Contextual Design moves quickly through those phases to get back to the users with our ideas represented as user interfaces.

Mock-up interviews using paper prototypes (Fig. 49.11) help designers understand why design elements work or fail and identify new function. These interviews are based on the principles of Contextual Inquiry given earlier. Test the paper prototype with users in their own context to keep them grounded in their real work practice. Users interact with the prototype by writing in their own content and manipulating and modifying the prototype. The partnership is one of co-design. As the user works with the prototype following a task s/he needs to do or

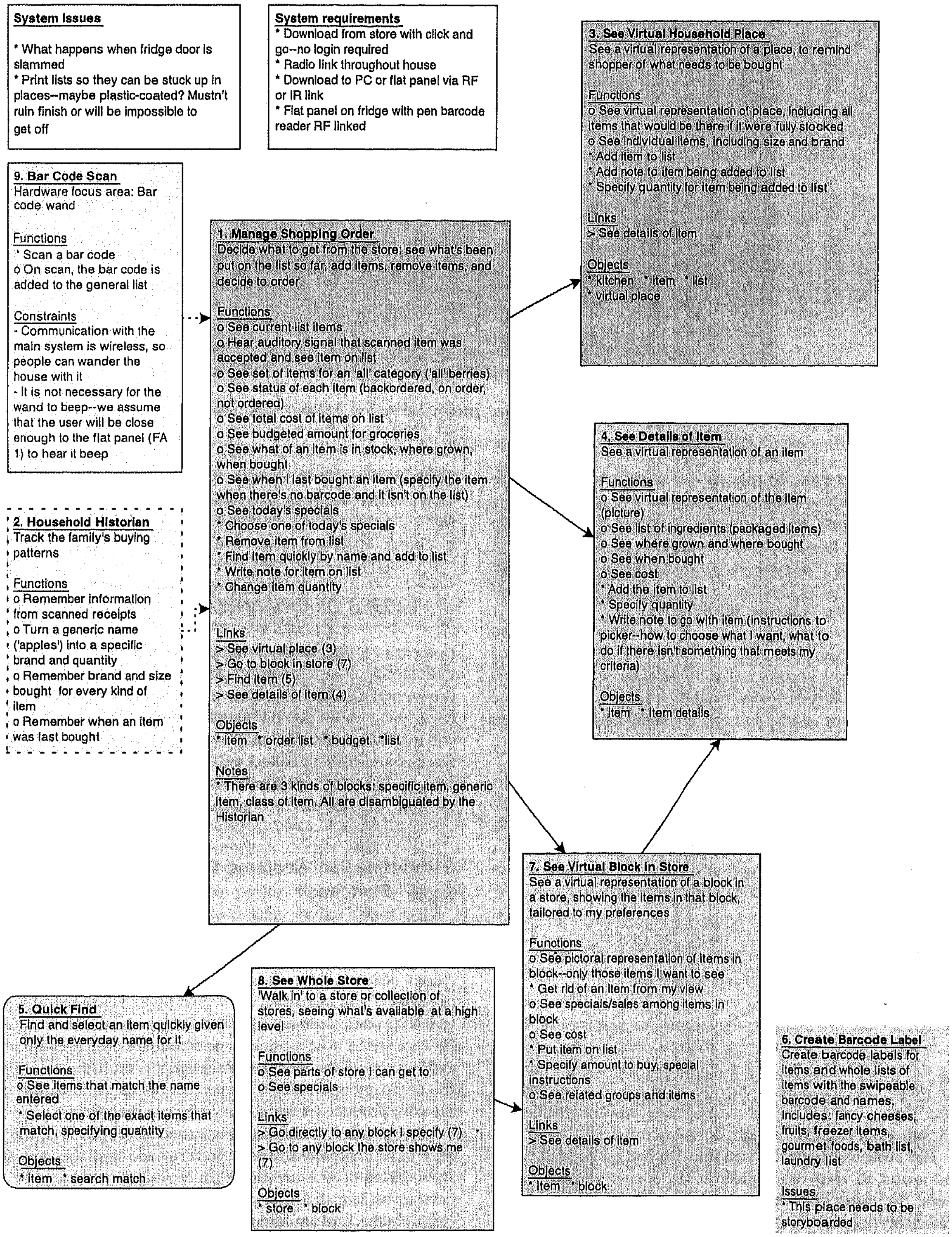

FIGURE 49.10. User Environment Design for a shopping system. IR = infrared; RF = radiofrequency.

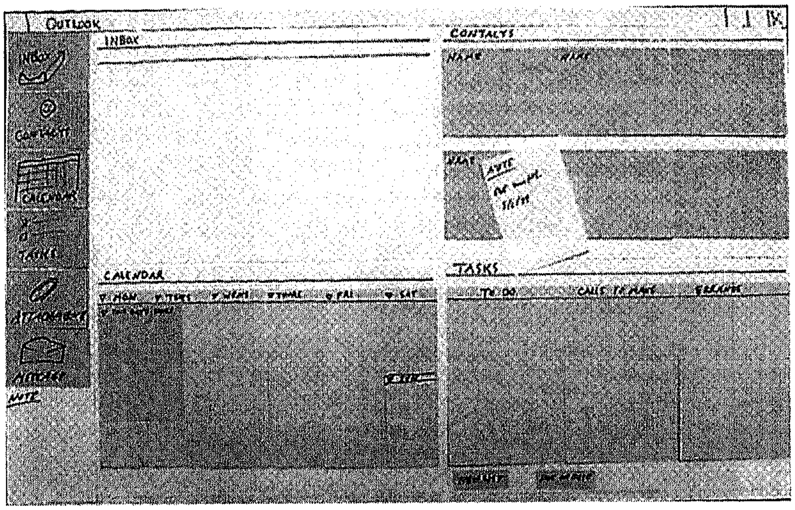

FIGURE 49.11. Build the paper prototype using normal stationary supplies.

did in the recent past, the user and designer uncover problems and adjust the prototype to fix them. Together the user and interviewer interpret what is going on in the usage and come up with alternative designs. Hand-drawn paper prototypes make it clear to the user that icons, layout, and other interface details are not the central purpose of the interviews; it keeps the user focused on testing structure and function.

After the design has been tested with between two and four users, redesign it to reflect the feedback. Multiple rounds of interviews and iterations allow testing in increasing levels of detail, first addressing structural issues, then user interface theme and layout issues, and finally detailed user interaction issues.

Let Customers Speed the Process. The goal of product design is to make something that people want and can use. If you only infuse customer data at the end of the process, you risk creating something that has a smooth interaction paradigm but does not support a valued part of the work. If you only infuse customer data at the beginning of the process, you risk having lots of good function that no one can access well. But, if you spend all your time in the middle of the process trying to get the function, layout, and user interface right before showing it to users, you risk overdesign and stagnation.

One company continued working on its User Environment Design for months because they could not resolve one issue or another, and they kept trying to get it perfect before returning to the field. Finally, their management was ready to cancel the project, which pushed them to go to the users and gather data. They talked to the users and the users resolved the issues, helping them identify the right design directions.

The customer is the bottom line. Do not get bogged down and spend all your time thinking. The sooner you get to the users, the faster your design will move forward. Even for a reasonably big project, all the steps of Contextual Design can be done in 2–3 months. A small project with limited models can be done in 5–6 weeks. After data consolidation, you should be out in the field within 2–3 weeks. Three rounds of mock-up interviews should be enough to stabilize the User Environment Design and validate the user interface theme and layout. If you are in a hurry, do three rounds with two users each instead of four.

ISSUES OF ORGANIZATIONAL ADOPTION

Contextual Design is about using customer data to produce the right design. Contextual Design is also about making a data-driven organization. If you have customer data, you can make decisions without getting bogged down in arguments and you will have a story for your market message. If you have processes that tell you how to collect and use customer data to produce designs, working together in cross-functional teams quickly is possible. But this means organizational and personal change.

Introducing User-Centered Design Is a Long Road—Start Small

Dennis Allen (1995), when he was at WordPerfect, talked about the water drop method of introducing Contextual Design. He would collect a little data and chat about it at lunch to key developers and managers. He would invite a few people to do an interpretation session. He arranged for a general interest talk. He collected a little more data and told more people. He made a slide show of the data. People started to want the data. Soon, the company was ready for a project. In time, 10 teams from the core product set were starting to use the process.

Change is first about changing attitudes. People have to know they have a problem. People have to see that the way they are working now is creating pain they need not have. People have to believe that it is possible to do better, that confused requirements and arguments over design do not have to come with the job. People have to have the bandwidth to try something new. People have to be willing to change.

Evangelists are the early adopters of new processes. They read about them and try to bring them into their companies. Building awareness through conversation, talks, and grass-roots activities always comes first. Culture and organizations change after their talk changes. Creating the buzz, creating a conversation about process, creating awareness about how people do their work, helps people understand why they should do something as onerous as changing their everyday habits.

This is true even when management is ready for the change. Now that user-centered design is seen as the "right way," managers are trying to change their organizations overnight. Our experience is that executives tend to expect results faster than is reasonable. CEOs often want everybody trained, user-centered processes implemented, and all the products delivered based on customer data immediately. This kind of management initiative can be very disruptive to the day-to-day work of making and selling products.

No matter who it is, grass-roots organizer or CEO, one person acting alone cannot change an organization's processes. Changing a whole organization requires effort at both the decision-making and grass-roots level: Spreading awareness and motivation; setting the new management direction; aligning people's work; changing evaluation criteria and milestone measures; training new skills; dealing with the people who are no good at the new skills; planning projects; tracking success and problems; changing the processes based on feedback; and making the new practice real in the context of a working project.

Change at a huge multinational company, with thousands of people busily working every day, will take years. It will be, like any large cultural movement, won on the ground: Person by person, project by project. So, evangelists: Give yourselves a break. Count every success no matter how small.

The best way to raise awareness is with the water drop technique. Do something small: Collect a little data, interpret it with a buddy, and share it around. Do guerrilla warfare: Find some others to join you, and collect a little data and interpret it without getting permission. Then share it around.

The next step is to identify a test project. Start with a project in an interested part of the company. Do not start with a big project or the most important product in the company. Make sure everyone is trained and set project scope small. Do a subset of the models. Use the data to drive design thinking—then perhaps go to your normal processes. Try the rest of the process on another project. Build success a bit at a time.

In this way, people can try parts of a new process on their own project, see the results, and share them. The Contextual Design process makes projects visible, because it requires a dedicated design room and creates many artifacts hanging on the wall. The environment it creates is so unusual that people frequently will visit the room just to see what is going on. Putting affinity diagrams on an outer wall will lead passersby into the room. A good design room, combined with the water drop method, will help create the buzz you need to get things going.

But remember, failure will kill it. One company had collected data from 100 people. They had taken months and had reams of data—observations cataloged on paper, none of which anyone used. But they had collected too much data, did not communicate it effectively, and did not get any of the results to affect a real shipping design. They lost management's confidence in trying a new process by producing no usable result. Now they will have to wait for new management, people who did not see this wasted work, and try to get them interested again. Pick something small and make sure it is successful. Know your techniques, manage your project scope, take care of each person's needs on the project, and do the painful leg work of organizing the project and getting customers yourself. Then publicize it throughout the company.

By combining upper level managers' pull with grass-roots evangelist push, the evangelist can ferret out pockets of interest inside the organization to initiate action. By setting reasonable expectations, these two parts of the organization will carry along the rest of the people who are engaged, heads-down, in their day-to-day work.

Communicate Out and Give Ideas Away

One team doing a Contextual Design project worried that the 600 coders they worked with would not adopt their designs. So, they identified and targeted friendly coders, asking them for help with prototyping, for their feedback on design ideas, and so forth. Next, in the interest of rolling out the process and the design, these designers gave ownership of the designs to the friendly coders, asking them to write the specifications, prototype the design, and give the prototypes to their managers. Soon, the coders were coming to them because they wanted the designs.

However, the designers did not want to just pass designs to the coders—that would not have achieved the overall awareness or driven customer-centered processes in the company. So, they started working only with coders who agreed to participate in field research, helping to collect and interpret customer work data. Even though this meant the designers gathered far more data than was necessary, they were willing to do this to spread awareness of the value of customer data throughout the company.

Doing something new often looks like a secret project. The team becomes involved with each other, talk a new language, and works a new way. If stakeholders, people on related projects, or the coders who will develop the designs think, "you act like you are *so* special" or if you keep telling them "how great this process is," they will resent it and resist changing. Communicating out and bringing people in is the way to be sure you do not become isolated:

- After you know what you are doing, bring other members of the team out with you on field interviews to take notes.
- Invite others into the interpretation sessions. Add one at a time so there are more people who know how to do the process (3 to 1 is good). It only takes two hours of their time, and then they will know what you are doing "all day in that room" and can help again in the future.
- Build the affinity as a big team event. Invite 10–15 people and involve them in building the data. Get their help in consolidating models, too. Pair each new person with someone on the team.

- Let interested parties and related teams walk the data and vision. Remember affinity wall walking and visioning are ways to hear their design ideas.
- Have bag lunches and open houses in the team room and tell people about your data. Collect their questions and challenges and use them to reset focus.
- Hang your data in the halls to pique interest and demonstrate openness.
- Publish on corporate Web sites and print newsletters to share your work across the organization. Editors and web masters are always looking for new stories.

Communicating out has side benefits as well. A large company had to hang their models up in the same hall that prospective customers for very large sales walked through. The company was able to demonstrate their understanding of the customer: Customers could identify themselves in the models!

Shipping product and changing organizations is all about getting buy-in from the people involved. Communicating, involving, and listening to people are part of the process. It has to be planned and managed by the people introducing the changes.

Ownership Is the Goal; Renovation Is Normal

All companies are already running themselves somehow. They may have very formal processes or they may be small start-ups with informal processes. They may have formal methodology groups in charge of their processes or they may have strong traditions that guide everyday work. Therefore, unless a founder who really wants to do Contextual Design starts a company, all introduction of Contextual Design involves fitting into existing ways of working. Any process, even if adopted by an enlightened CEO with an enlightened set of developers, will have to be changed to fit the company, the people, and their skills. Any user-centered design process, will be renovated and adopted piece by piece.

Part of any adoption process is to make it your own. People will argue that they must change a process to fit their situation or to improve it. Early on in the evolution of Contextual Design, there was a step called "redesigned sequence models," a rewrite of the consolidated sequence steps to reflect the vision. An internal evangelist could not get anyone to use redesigned sequence models. Instead, their people were excited about storyboarding as a technique, which the evangelist realized was just redesigned sequence models but in drawings. So they did storyboarding and were happy. Renaming or modifying a step in Contextual Design makes no difference, as long as the fundamental intents of the step are achieved. Because of the value of visualizing the steps pictorially, storyboarding was incorporated into the Contextual Design process.

When adopting a new process, you cannot make change happen faster than people's awareness, growing skill, and willingness to change. People will resist a new process until they can find a place for themselves within the process and a place for the process within their organization. They have to figure this out themselves.

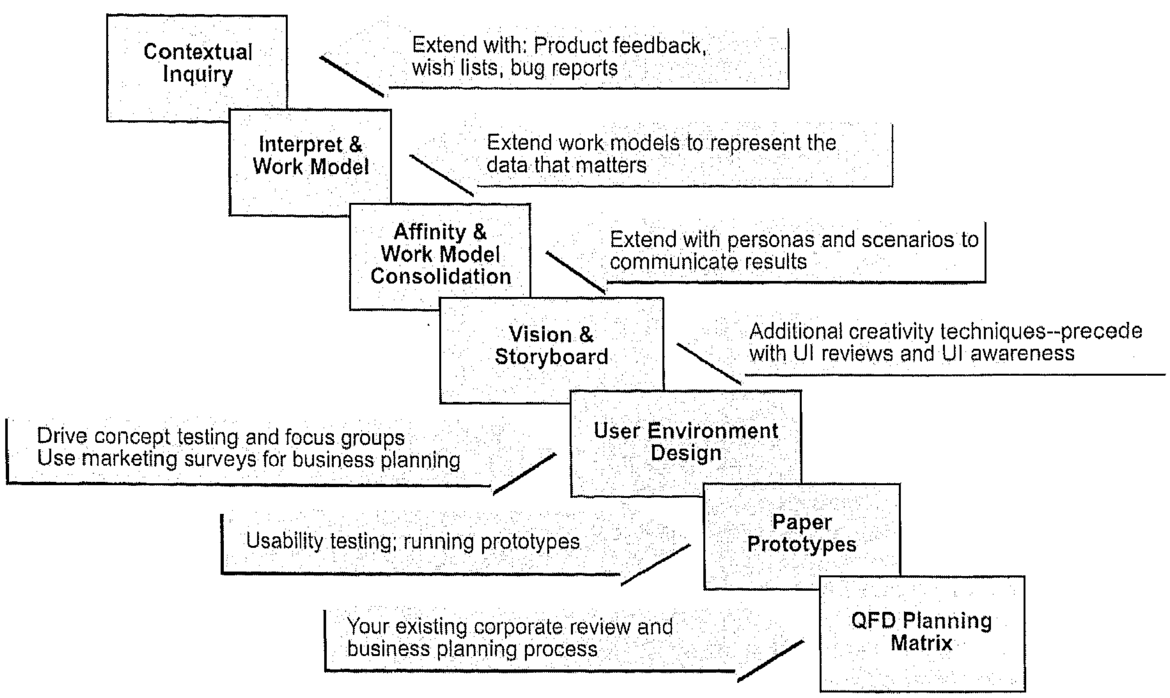

FIGURE 49.12. The steps of Contextual Design act as a process scaffolding upon which complementary activities can be included. QFD = Quality Function Deployment; UI = User Interface.

One company was very excited about collecting customer data, and they started sending hundreds of people out into the field. Soon they were overwhelmed with the amount of data and started looking for a way to handle it. They tried everything—individual reports, group presentations, and postinterviewing data dumps—and still could not handle the data or get people to really think about it. Then their Contextual Design evangelist suggested, "What if we tried this technique called an interpretation session, where you write down all the points and model the data?"

That is simply the second step in Contextual Design; but, until they had tried their own ideas, they could not believe that the interpretation session was efficient. Companies may resist each step of Contextual Design, and then adopt each piece with slight tuning. Companies may lose faith in the process because of an early failure—yet, years later, they are still doing key techniques under different names. There are lots of companies and designers who are trying particular Contextual Design techniques in many diverse projects.

It is not a failure of adoption if people resist or change a particular step—it is part of their adoption process. Training, coaching, and advising will always initiate the adoption process, but it is the people working with it on a daily basis who have moved the industry forward over the last 13 years. These are the people who change and tune the process to make themselves and their organizations comfortable. These people have made customer-centered design and Contextual Design a part of the daily work of companies all around the globe.

Contextual Design is a set of steps (Fig. 49.12) that create scaffolding that organizations can use as a base to build their own approach to understanding the customer and getting that data into their designs. To change, people need to know exactly what to do to be successful. They need to know the techniques, how to work together, and how to fit this work into their daily life. Therefore, change looks like taking on one piece or technique at a time. Most companies start by going to the field to gather customer data and proceed from there. Each step is a success. Each success must be recognized and celebrated.

Looking back, the industry has come a long way. Today, companies all over the world are designing from customer data. The argument is now what customer data to collect and how to collect it, not whether there should be customer data. Using customer data is now a central part of the corporate dialogue on how to make products. I am pleased that Contextual Design has been a part of this revolution.

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